

```

import pandas as pd
import numpy as np

data = pd.read_csv('knn_brightness_saturation_9_6_2.csv')

print(data.shape)
print(data.head())

(7, 3)
  Brightness  Saturation Class
0          40          20   Red
1          50          50  Blue
2          60          90  Blue
3          10          25   Red
4          70          70  Blue

data.isnull().sum()

Brightness    0
Saturation    0
Class         0
dtype: int64

print("Existing Dataset:")
print(data)

Existing Dataset:
  Brightness  Saturation Class
0          40          20   Red
1          50          50  Blue
2          60          90  Blue
3          10          25   Red
4          70          70  Blue
5          60          10   Red
6          25          80  Blue

new_brightness = 45
new_saturation = 40

print("New Entry:")
print("Brightness:", new_brightness)
print("Saturation:", new_saturation)

New Entry:
Brightness: 45
Saturation: 40

data['Euclidean_Distance'] = np.sqrt(
    (data['Brightness'] - new_brightness) ** 2 +
    (data['Saturation'] - new_saturation) ** 2
)

```

```
print(data)
```

	Brightness	Saturation	Class	Euclidean_Distance
0	40	20	Red	20.615528
1	50	50	Blue	11.180340
2	60	90	Blue	52.201533
3	10	25	Red	38.078866
4	70	70	Blue	39.051248
5	60	10	Red	33.541020
6	25	80	Blue	44.721360

```
print("Euclidean Distance from New Entry:\n")
```

```
for i in range(len(data)):
    print(
        "Point", i + 1,
        ":",
        "Brightness =", data.loc[i, 'Brightness'],
        ", Saturation =", data.loc[i, 'Saturation'],
        ", Class =", data.loc[i, 'Class'],
        ", Distance =", round(
            data.loc[i, 'Euclidean_Distance'], 3
        )
    )
```

```
Euclidean Distance from New Entry:
```

```
Point 1 : Brightness = 40 , Saturation = 20 , Class = Red , Distance =
20.616
Point 2 : Brightness = 50 , Saturation = 50 , Class = Blue , Distance
= 11.18
Point 3 : Brightness = 60 , Saturation = 90 , Class = Blue , Distance
= 52.202
Point 4 : Brightness = 10 , Saturation = 25 , Class = Red , Distance =
38.079
Point 5 : Brightness = 70 , Saturation = 70 , Class = Blue , Distance
= 39.051
Point 6 : Brightness = 60 , Saturation = 10 , Class = Red , Distance =
33.541
Point 7 : Brightness = 25 , Saturation = 80 , Class = Blue , Distance
= 44.721
```

```
sorted_data = data.sort_values(
    by='Euclidean_Distance'
)
```

```
print("Sorted Data:")
```

```
print(sorted_data)
```

```
Sorted Data:
```

	Brightness	Saturation	Class	Euclidean_Distance
1	50	50	Blue	11.180340
0	40	20	Red	20.615528
5	60	10	Red	33.541020
3	10	25	Red	38.078866
4	70	70	Blue	39.051248
6	25	80	Blue	44.721360
2	60	90	Blue	52.201533

```
data.to_csv(
    'knn_euclidean_distances_9_6_2.csv',
    index=False
)

print("Distance CSV file created successfully!")

Distance CSV file created successfully!

import matplotlib.pyplot as plt

red_data = data[data['Class'] == 'Red']
blue_data = data[data['Class'] == 'Blue']

plt.scatter(
    red_data['Brightness'],
    red_data['Saturation'],
    label='Red'
)

plt.scatter(
    blue_data['Brightness'],
    blue_data['Saturation'],
    label='Blue'
)

plt.xlabel('Brightness')
plt.ylabel('Saturation')
plt.title('Brightness vs Saturation')
plt.legend()
plt.grid(True)

plt.show()
```

